

Ingrid R. Carrasco

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Education

- Florida Institute of Technology**, BS in Planetary Sciences August 2022 – May 2026
- **Coursework:** Observational Astronomy, Modern Physics, Electromagnetic Theory, Quantum Mechanics, Partial Differential Equations, Remote Sensing, Physical Mechanics, Geophysics, Astrophysics, & Instrumentation.
- Oregon State University**, Salem, Oregon June 2024 – August 2026
- **Coursework:** Differential Equations and Linear Algebra
- Chemeketa Community College**, Salem, Oregon Sept. 2020 – August 2022

Research Experience

- Exoplanet Habitability Evolution**, FIT – Melbourne, FL August 2025 – May 2026
- Modeled the long-term habitability of the icy satellite Enceladus, focusing on how the dramatic changes in stellar radiation from the Sun's Red Giant and White Dwarf evolution affect its subsurface environment.
 - Utilized the ExoPlaSim General Circulation Model (GCM) to simulate climate and atmospheric conditions, assessing the potential for a temporary enhancement in habitability during the post-main-sequence era.
- CERN CMS Upgrade & Testing**, FIT – Melbourne, FL January 2025 – May 2026
- Tested and graded semiconductor chips for the CMS Inner Tracker upgrade to ensure performance standards for high-energy particle detection in the High-Luminosity LHC (HL-LHC)
 - Investigated anisotropic resistance in carbon fiber materials to determine conductivity variations through building a testing apparatus.
- Lightning Detection Spectroscopy for Exoplanets**, UC Berkeley – Berkeley, CA June 2025 – July 2025
- Hands-on build of a complete spectrograph system, including optical alignment, calibration, and instrumentation setup.
 - Spectroscopic measurements performed to detect exoplanetary atmospheric signatures.
- Plant the Moon & Mars Competition**, FIT – Melbourne, FL August 2024 – May 2025
- Led research team evaluating soil amendments and plant growth under simulated Martian conditions
 - Designed experiments, optimized organic material compositions, maintained environmental controls, and ensured precise data collection.

Leadership Experience

- Ortega Observatory Assistant**, FIT – Melbourne, FL August 2025 – May 2026
- Training in telescope operation and advanced observational techniques, including handling multiple instruments (such as Unistellar and Celestron systems) for night sky acquisition and data collection.
 - Assisting with public outreach events at the observatory and Kennedy Space Center, guiding visitors through celestial observations, and promoting science education.
- Astronomical Research and Education Society**, President – Melbourne, FL August 2022 – October 2025
- Lead and oversee ARES, a student-led research organization with more than 10 active projects in engineering, astronomy, astrobiology, and software development, fostering innovation and collaboration.
 - Organize guest speaker events featuring astronauts and scientists, coordinate study nights, and support student research initiatives to enhance academic and professional development in space sciences.
- Gamma Phi Beta International Sorority**, Officer – Melbourne, FL August 2022 – October 2025
- Fidelity Chairwoman, created presentations to educate members on various topics including career development, STEM pathways, inclusivity/diversity, and empowering womanhood.

Technical Skills

Programming & Software: C++ (Intermediate), Python (Intermediate), JavaScript (Beginner), Ubuntu (Expert), Google Colab (Expert), Visual Studio (Expert)

Data Analysis: Spectroscopy, Remote Sensing, Astronomical Data Reduction

Instrumentation and Observing: Telescope Operation (Celestron, Unistellar, Meade), Optomechanical Systems

Membership

American Physical Society Student Member: 2022 - Present

Gamma Phi Beta Fidelity Officer: 2024 - Present

Astronomical Research and Engineering Society President: 2022 - 2025

American Meteorological Society member: 2022 - 2025

Service

Letters to A Pre-Scientist: (2022 - Present) Encouraging STEM interest in elementary school students through writing letters about my personal experience.

Princesses With Power Tools: (2023-2025) Breaking barriers for women in STEM by teaching young girls how to use power tools.

Languages

English: Speaking (Native), Writing (Native), Reading (Native)

Spanish: Speaking (Native), Writing (Native), Reading (Native)

German: Speaking (Beginner), Writing (Beginner), Reading (Intermediate)

References

Dr. Souvik Das: Professor at Florida Institute of Technology, [sdas@fit.edu].

Dr. Howard Chen: Professor at Florida Institute of Technology, [hchen@fit.edu]

Dr. Luis Quiroga-Núñez: Professor at Florida Institute of Technology and Director of the Ortega Observatory, [lquiroganunez@fit.edu] (Additionally serving as my supervisor at Ortega Observatory)